SOP-KNOWLEDGE-INGESTION-005: Admin Document Injection for Hybrid RAG

Version 1.0 | Effective 2026-03-10 | Classification: Synthetic / Non-Clinical

1. Purpose

Describe how Knowledge Admins inject fictional SOPs and digests into the local BM25 index and vector store.

2. Supported Formats

TXT, PDF, DOCX, and CSV metadata may be used in this training prototype.

All content must remain synthetic.

3. Ingestion Steps

Step 1: Open Admin Document Injection page in Streamlit.

Step 2: Upload a fictional SOP (for example SOP-LOT-QUARANTINE-001).

Step 3: System chunks text using approximately 512-token windows with 64-token overlap.

Step 4: Bi-encoder embeddings are written to the local vector store.

Step 5: The same chunks are added to the BM25 lexical index.

Step 6: Audit log records document_ingested.

4. Retrieval Reminder

Queries use BM25 + dense retrieval + RRF (k=60) + cross-encoder rerank (drop below 0.40).

Only top 3 to 5 chunks are passed to generation.